

VOICE CLONING

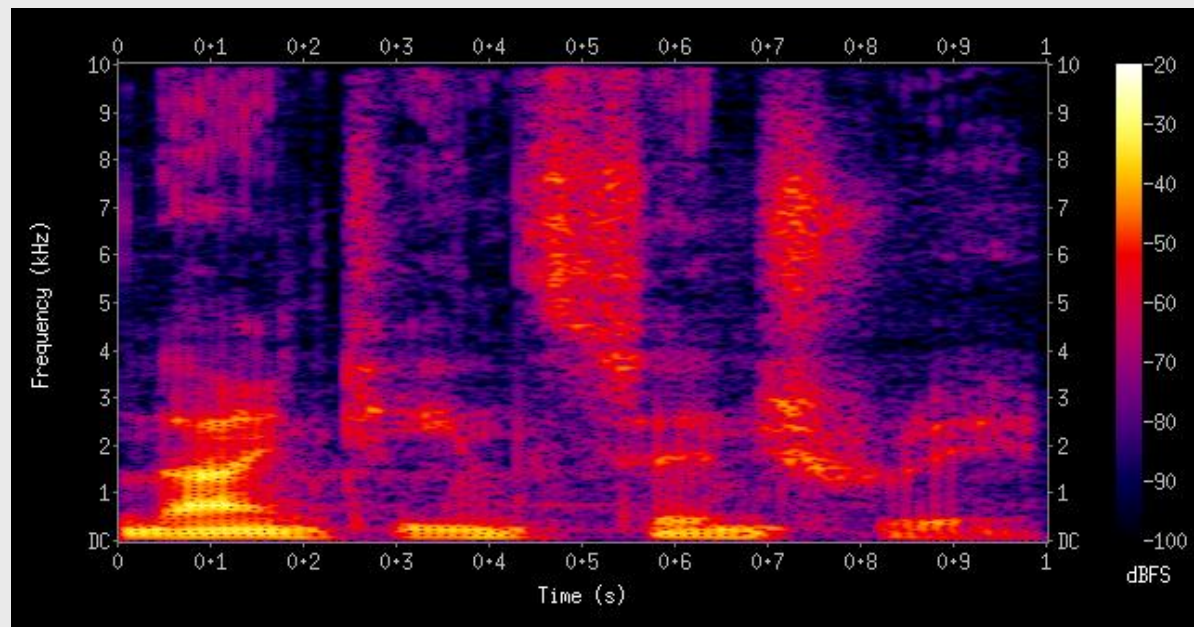


AI Voice Cloning

- AI can easily mimic the ***text*** of someone's speaking style with transformers
- This is possible because the AI is able to represent text with a good embedding vector
- If the AI wanted mimic the ***sound*** of someone's speaking style, what would the embedding vector represent?

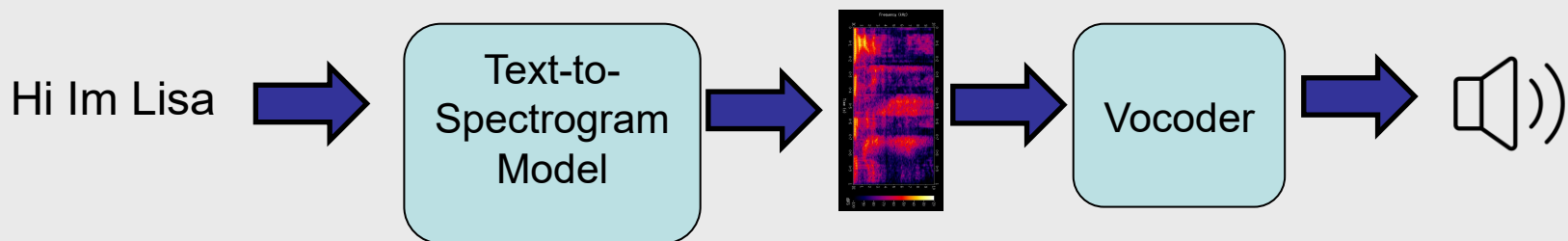
Spectrograms: How Computers Represent Voices

- Spectrogram – represents the frequencies in an audio signal as it varies in time
 - Like an audio fingerprint



Text-to-Speech (TTS) Models

- Text to speech (TTS) models are neural networks that convert text into spectrograms
- The spectrograms are then turned into voice audio using neural networks called vocoders



First Good TTS Model: Tacotron

TACOTRON: TOWARDS END-TO-END SPEECH SYNTHESIS

Yuxuan Wang^{*}, RJ Skerry-Ryan^{*}, Daisy Stanton, Yonghui Wu, Ron J. Weiss[†], Navdeep Jaitly,

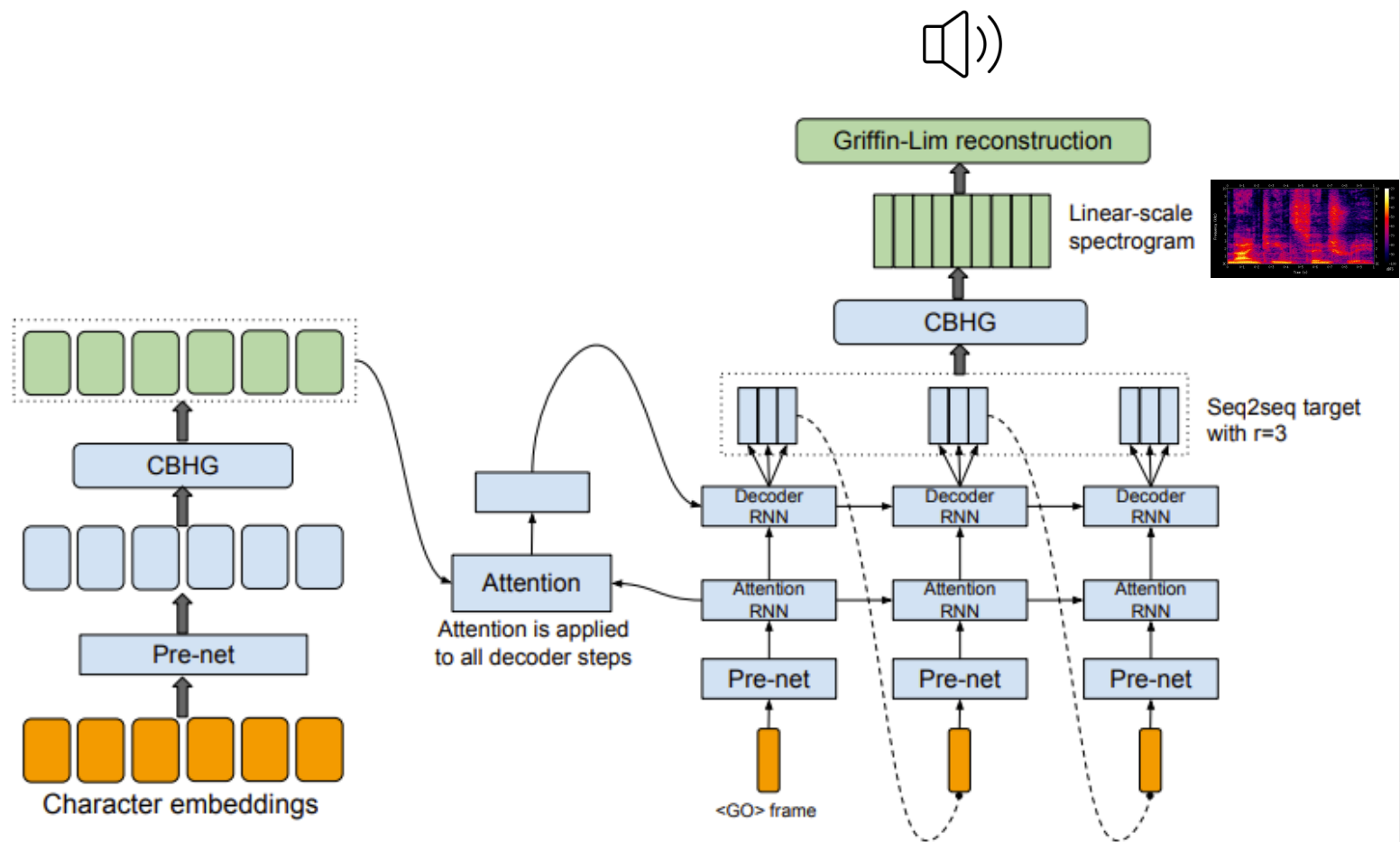
Zongheng Yang, Ying Xiao^{*}, Zhifeng Chen, Samy Bengio[†], Quoc Le, Yannis Agiomyrgiannakis,

Rob Clark, Rif A. Saurous^{*}

Google, Inc.

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Tacotron Architecture



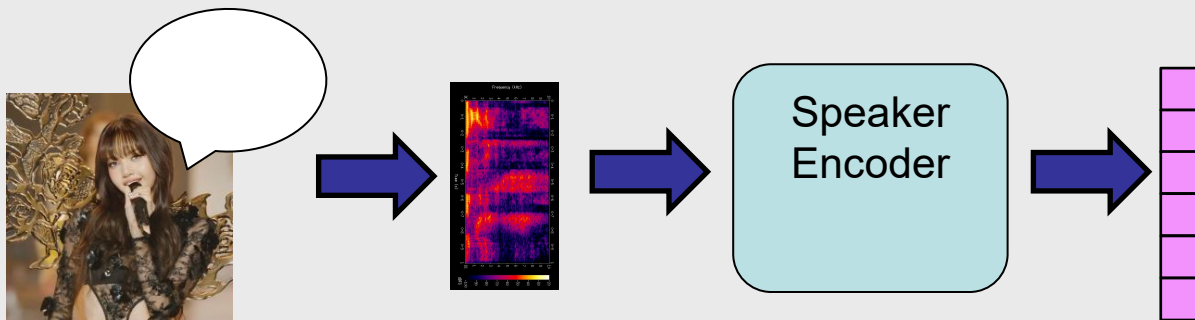
Hi Im Lisa

Modern TTS

- **VALL-E (2023)**
 - Uses a transformer to generate audio codec tokens
 - Input is text + audio sample of voice you want
- **Voicebox (2023)**
 - Use a diffusion model to create the audio
 - Drift velocity is a transformer
 - Input is text + audio sample of voice you want

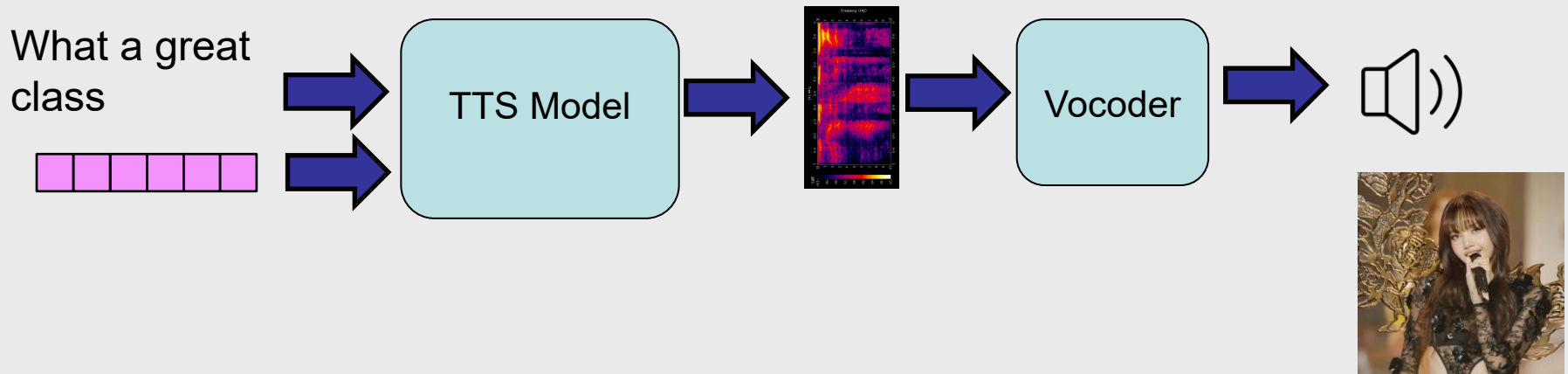
Representing a Speaker

- Spectrograms are like voice fingerprints
- Voice cloning neural networks called “speaker encoders” take spectrograms and turn them into vectors
- Trained using contrastive learning



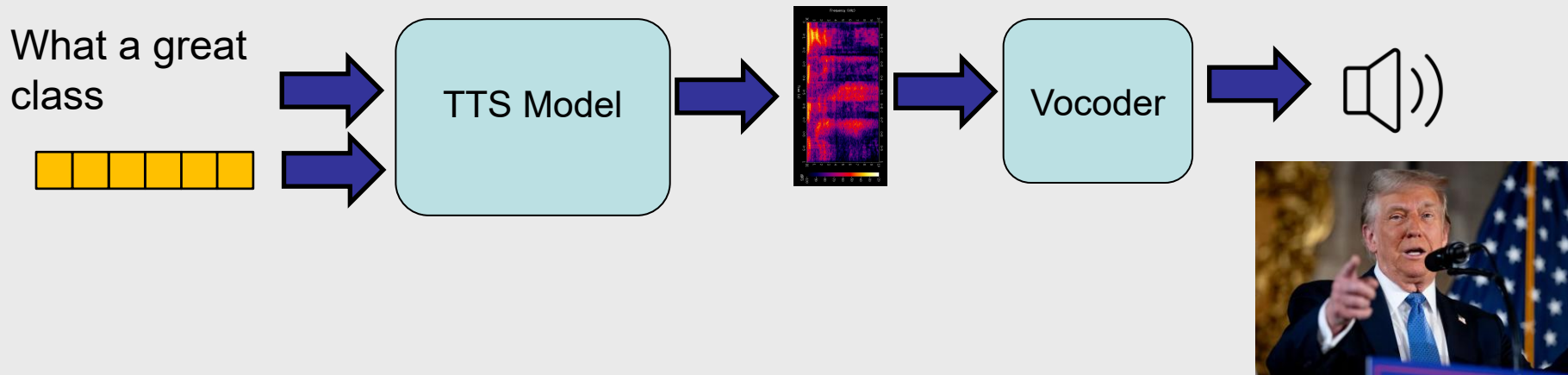
Voice Cloning

- Once we have the speaker vector, we can just add it to the TTS model



Voice Cloning

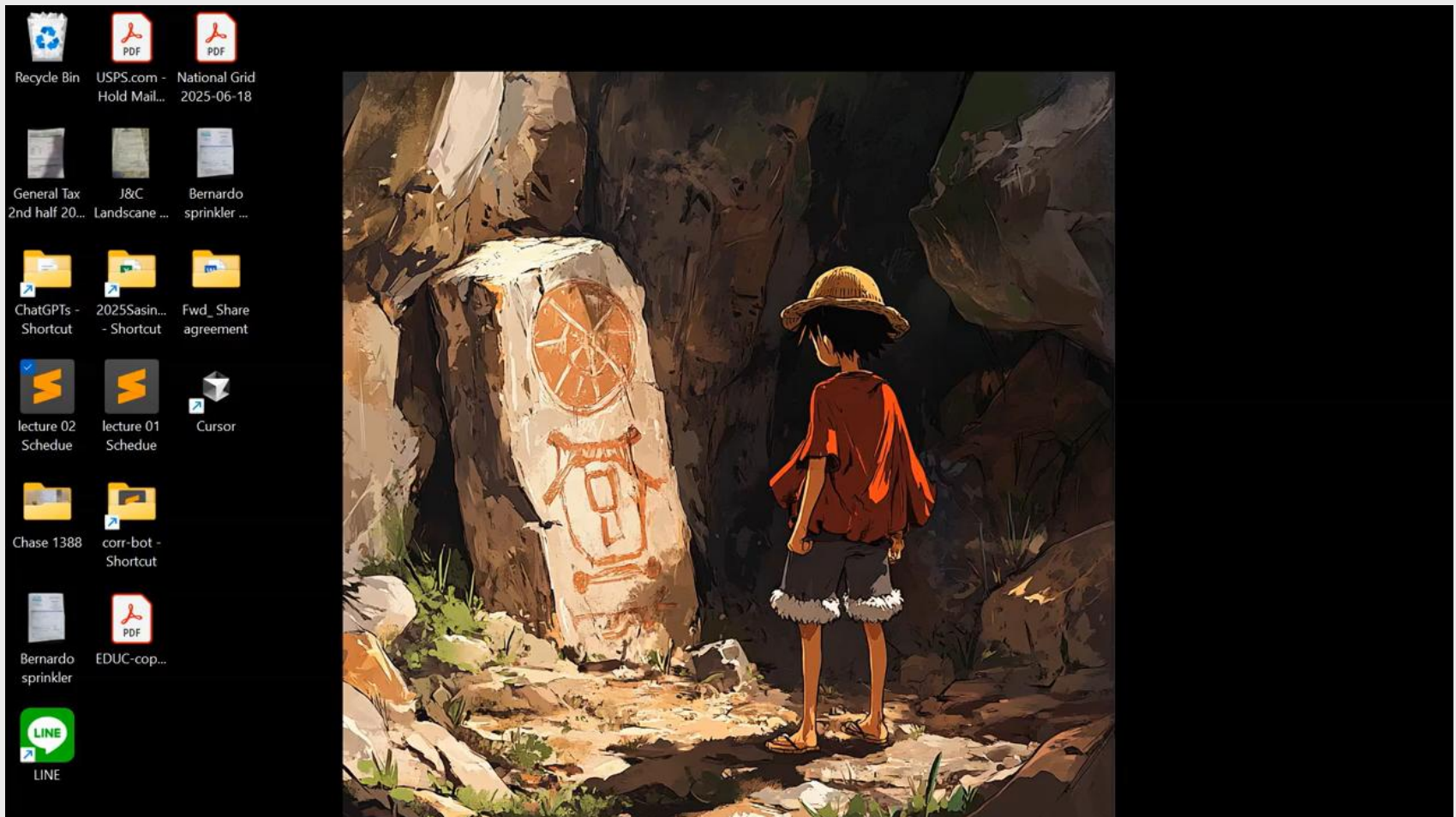
- Once we have the speaker vector, we can just add it to the TTS model



ElevenLabs

- **ElevenLabs is one of the most advanced voice cloning tools: <https://elevenlabs.io/>**
- **Many voice cloning features**
 - Voice library (use voices others cloned)
 - Instant voice cloning (10 seconds of audio)
 - Professional voice cloning (30 minutes of audio)
 - Voice design (create voice by text prompt)
 - Voice changer (change the voice in an audio file)

Voice Changer



Permission Granted to Voice Clone

IIElevenLabs

Iconic Voices



Permission Granted to Voice Clone

James Earl Jones Signed Over Rights For AI To Recreate Darth Vader's Voice

By [Tim Lammers](#), Contributor. © I cover Hollywood and entertainment.

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Permission Not Granted to Voice Clone

Robert Downey Jr. vows there will never be a digital AI replica of him on-screen

News

By Eric Hal Schwartz published October 31, 2024

Ultron isn't Iron Man's only AI foe



Voice Cloning Ethics

- **If using a voice clone, be transparent**
- **Don't use anyone's voice without their permission**
- **Don't make the voice clone say offensive things**
- **Don't create deepfake audio that will cause panic...**

How to Clone a Voice

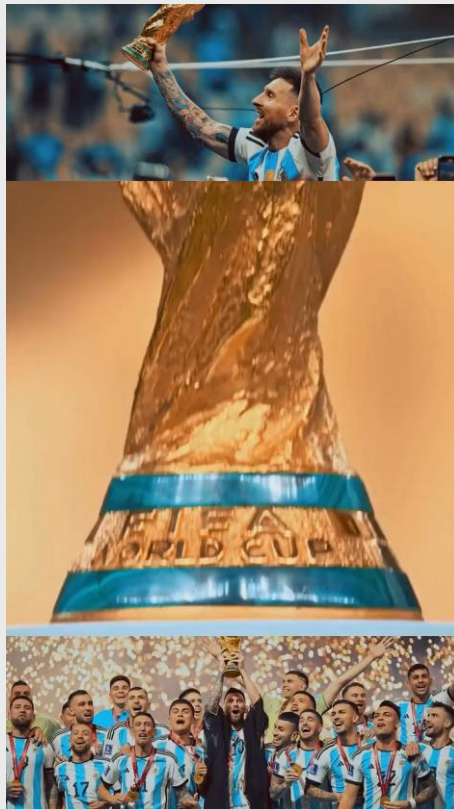
- **ElevenLabs allows us to clone voices with only a few seconds of audio**
- **Can obtain audio files from YouTube**
 - **Choose a YouTube video of the subject speaking alone or mostly alone**
 - **Download audio of YouTube video**
(<https://tuberipper.net/73/>)
 - **Clip the audio to 10 to 30 second clips**
(<https://audiotrimmer.com/>)

Video Narrations

- **If we give the frames of a video to AI plus some instructions, it can give us a narration for the video**
- **We already have methods to gives multiple images and context text to AI and get a text description**
- **We just need to extract the images from the video**
 - **We will usually sample around 10-20 images per video**
 - **This is near the limit of what most models allow**

Old HW Problem

- **Make a narration of this video**
- **Bonus – clone a voice and have it narrate the video**



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